



KCL AND KVL

CIRCUIT ANALYSIS METHOD

prepared by:

Gyro A. Madrona
Electronics Engineer

TOPIC OUTLINE

Circuit Convention

Kirchhoff's Current Law (KCL)

Kirchhoff's Voltage Law (KVL)



CIRCUIT CONVENTION



CONVENTION

A convention is a widely accepted practice, method, or behavior that is followed by common agreement or tradition, rather than by formal rules.

Color coding in Offices

red – urgent documents

blue – general files

green – financial records

This is a common practice but not formally regulated.



STANDARD

A standard is a formal, established guideline, rule, or specification that is often mandatory and enforced by an authoritative body or organization.

IEC 60062 Resistor Color Code

black – 0

brown – 1

red – 2

.

.

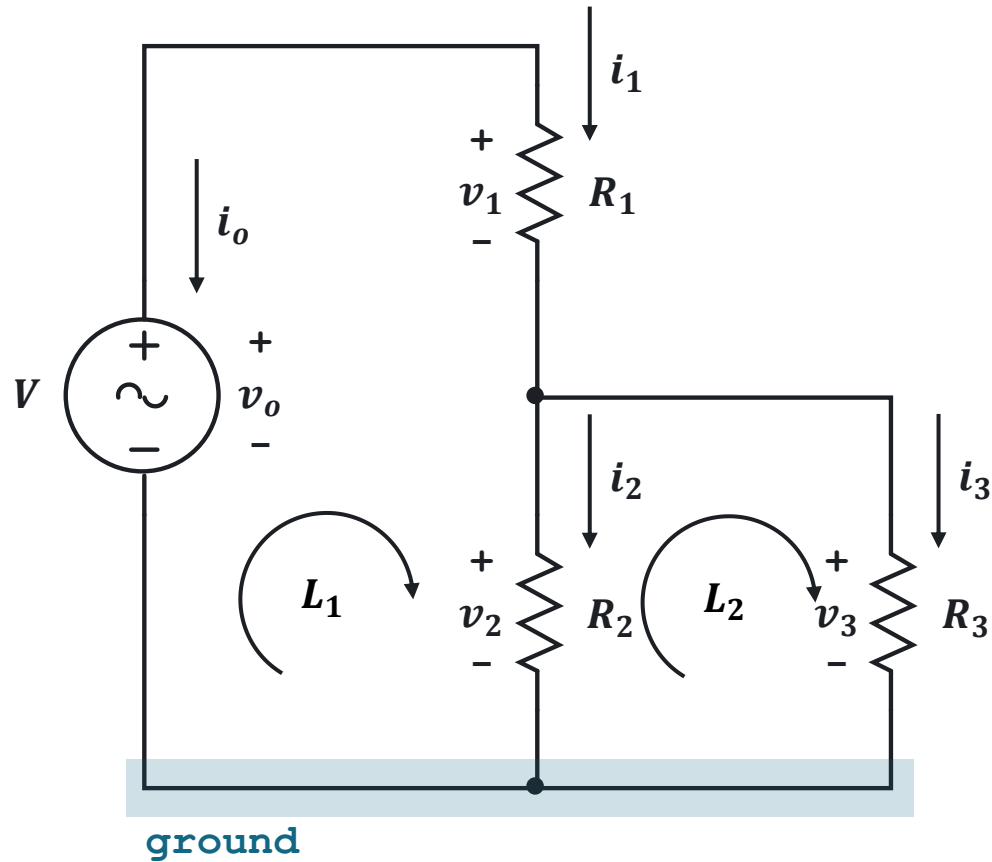
.

white – 9

Resistors have colored bands that represent specific digits, multipliers, and tolerance values.



LABELING VARIABLES



Steps in Labeling Variables

1. Label the Reference Node (ground):

Select a reference node with the most connections or the negative (-) terminal of a voltage source.

2. Label Node Voltages:

Mark higher potentials as positive (+) relative to the reference node.

3. Label Currents:

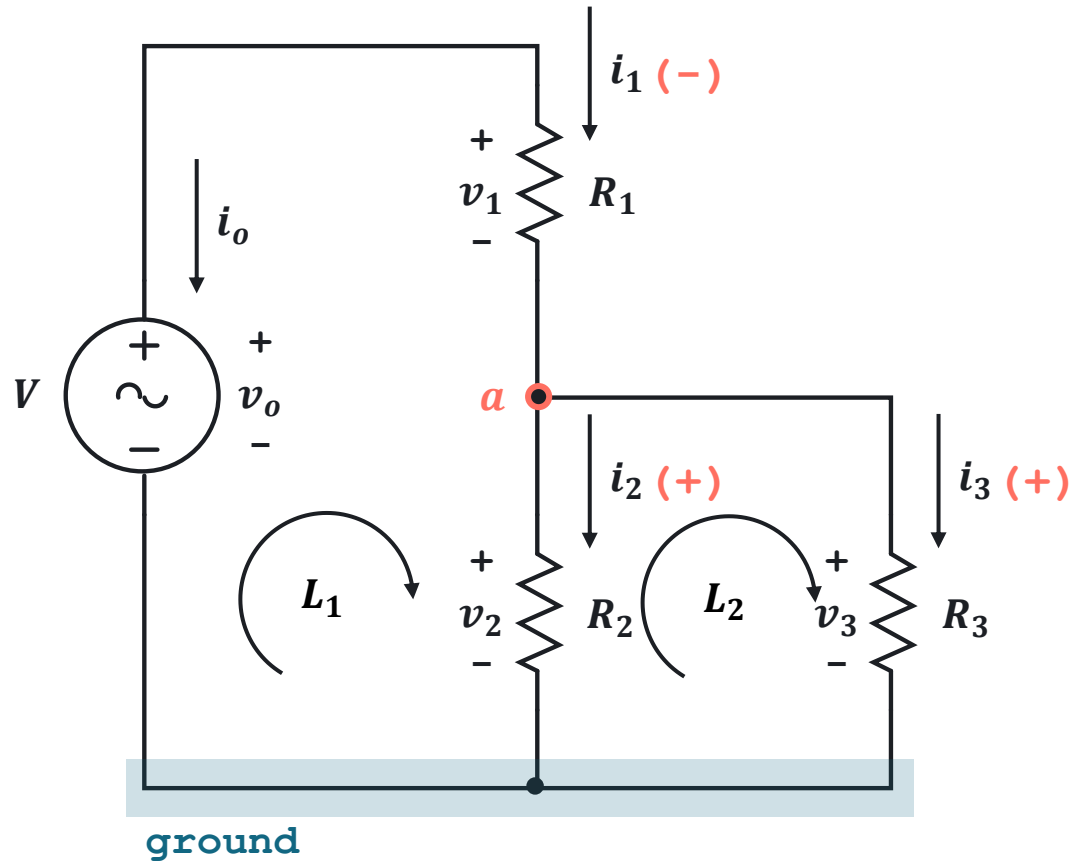
Entering the positive (+) terminal of a component.

4. Create a voltage loop:

Follow the defined current directions.



CURRENT FLOW CONVENTION



Current Flow Convention

- Current entering a node is negative $(-)$
- Current leaving a node is positive $(+)$

@ a

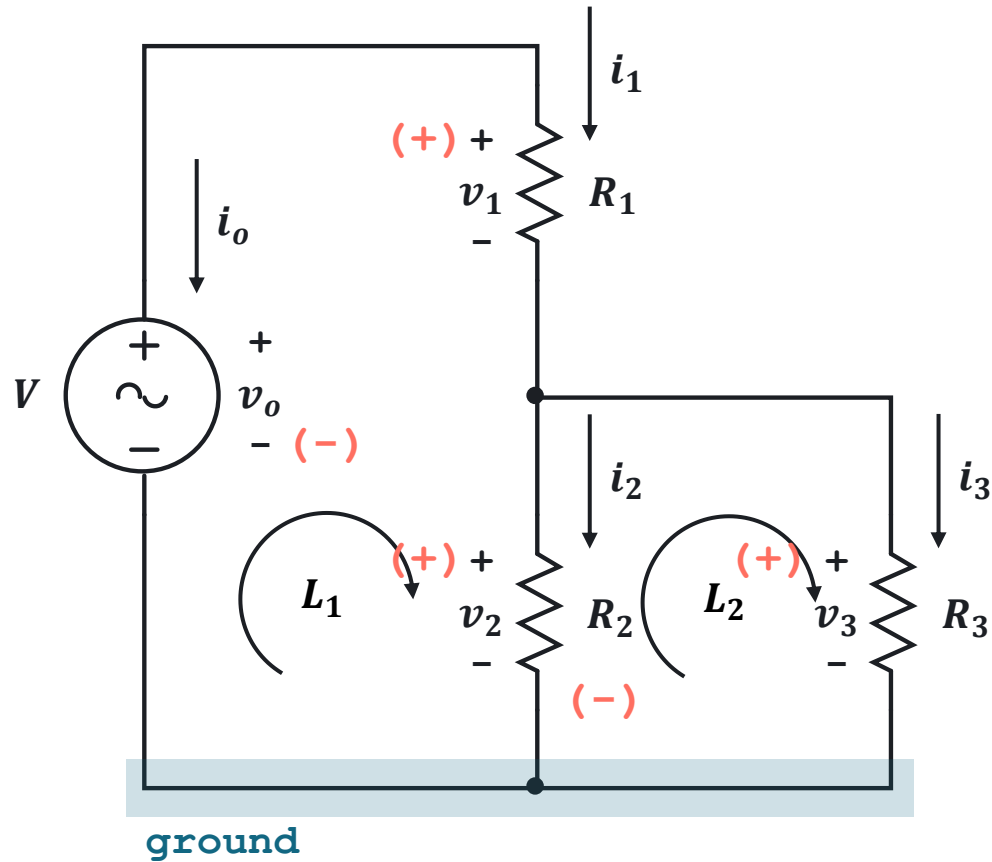
$-i_1$

$+i_2$

$+i_3$



VOLTAGE LOOP CONVENTION



Voltage Loop Convention

The “sign” of voltage of the element is the first sign the loop encounters.

@ L_1

$-v_0$

$+v_1$

$+v_2$

@ L_2

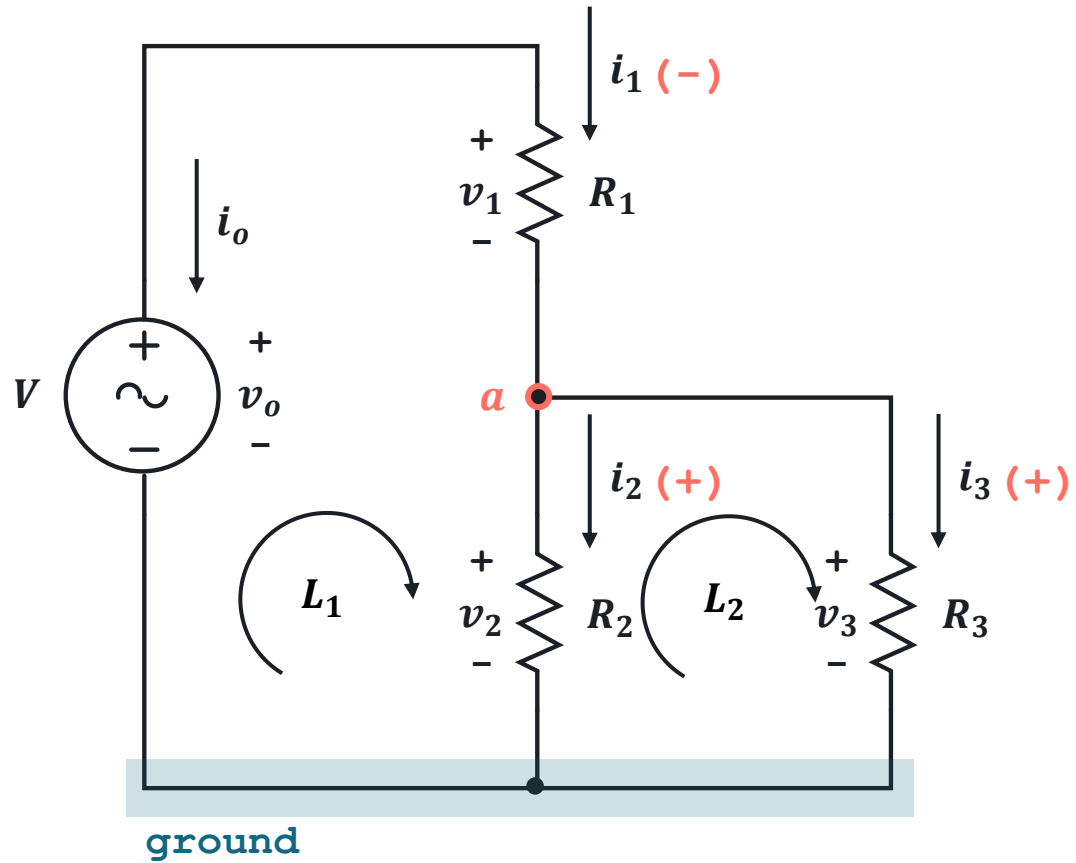
$-v_2$

$+v_3$



KIRCHHOFF'S CURRENT LAW AND VOLTAGE LAW





Kirchhoff's Current Law

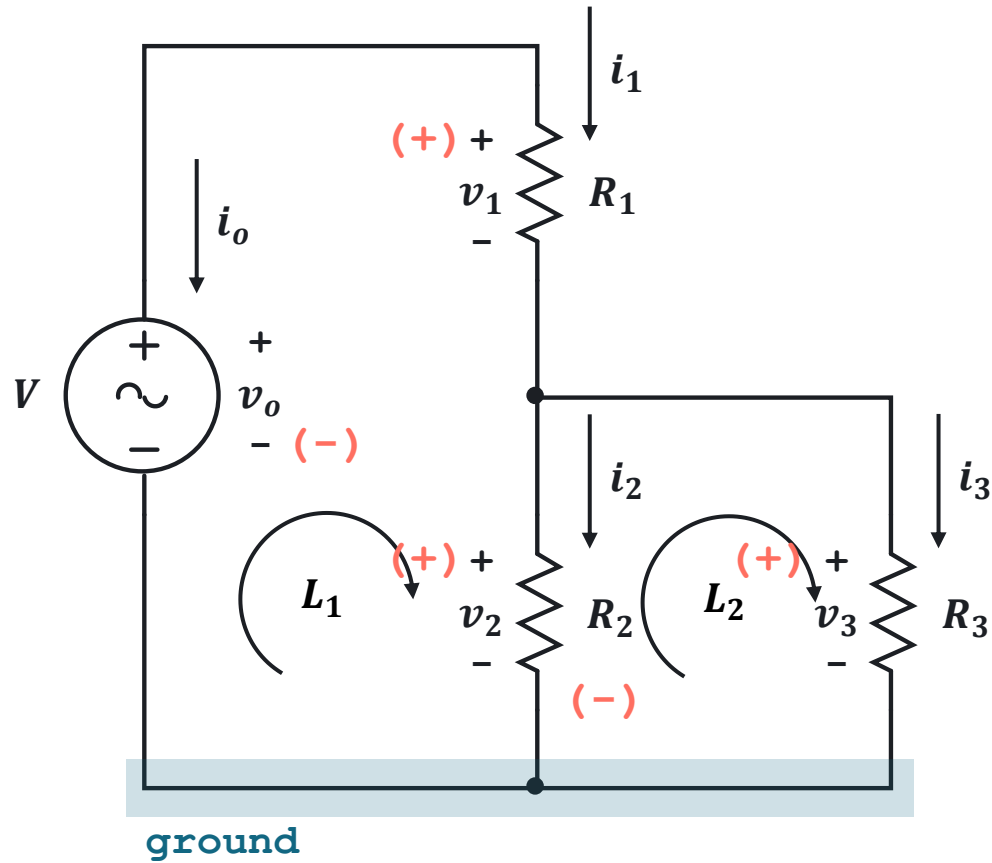
The summation of currents going-in and going-out a node is zero.

$$\sum i_j = 0$$

KCL @ a

$$-i_1 + i_2 + i_3 = 0$$





Kirchhoff's Voltage Law

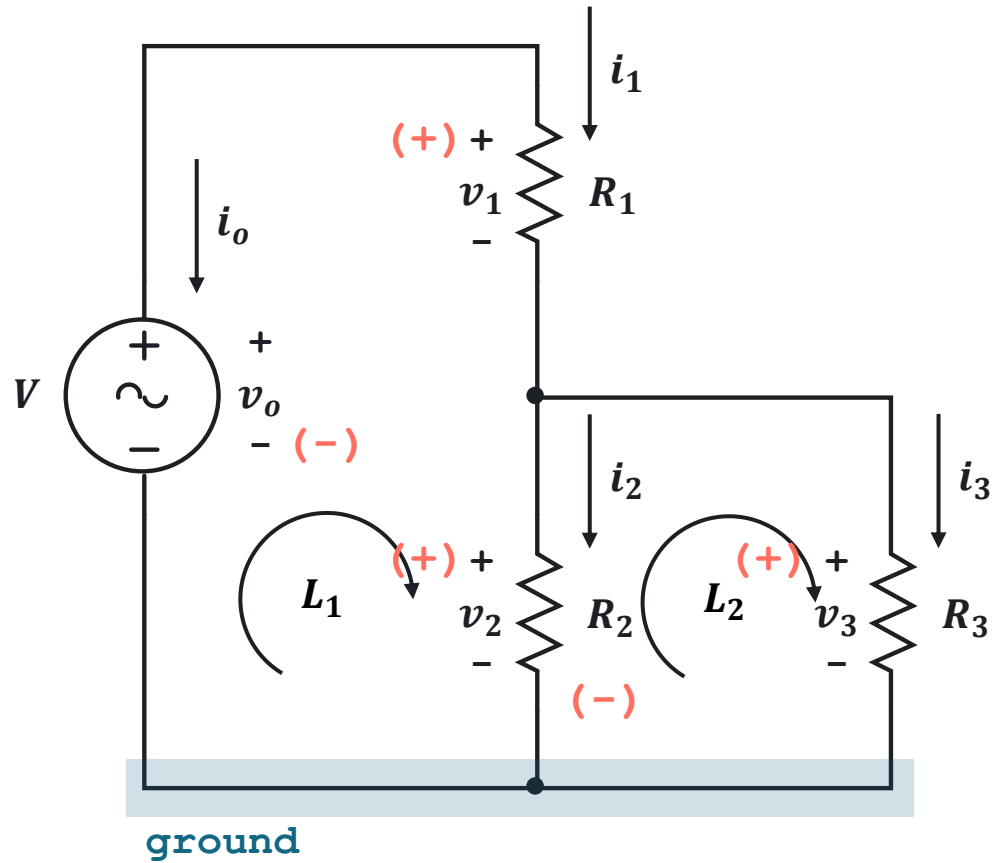
The summation of voltages in a closed-loop is zero.

$$\sum v_j = 0$$

KVL @ L_1

$$-v_0 + v_1 + v_2 = 0$$





Kirchhoff's Voltage Law

The summation of voltages in a closed-loop is zero.

$$\sum v_j = 0$$

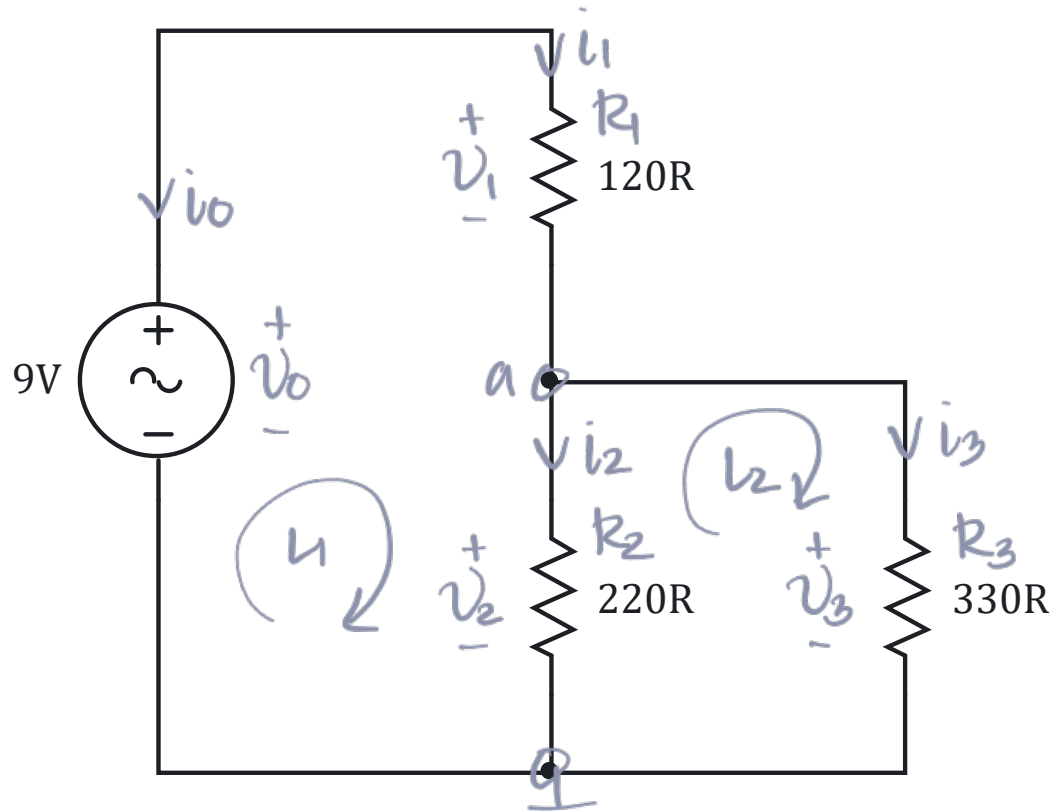
KVL @ L_2

$$-v_2 + v_3 = 0$$



EXERCISE

Determine the **current** flowing through each resistor and the **voltage** drop across each resistor in the given circuit.



Solution

KCL @ a

$$-i_1 + i_2 + i_3 = 0$$

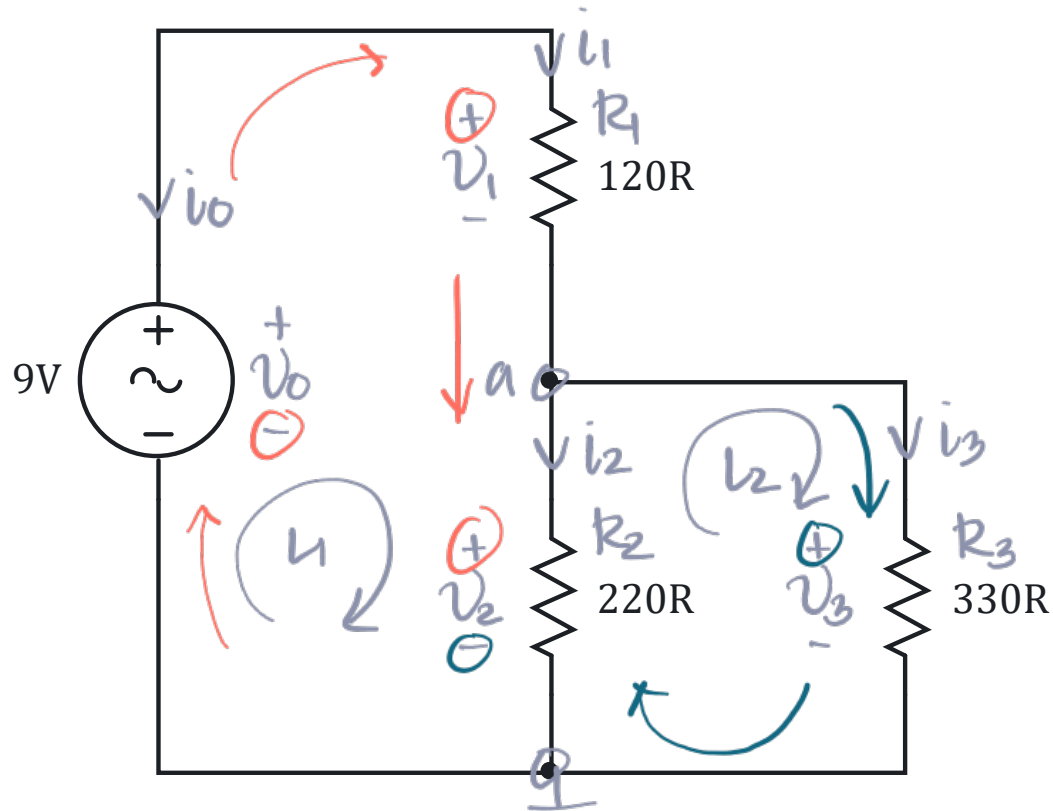
$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} = 0$$

$$120 \left[-\frac{1}{120} v_1 + \frac{1}{220} v_2 + \frac{1}{330} v_3 \right] = 0$$

$$-v_1 + \frac{6}{11} v_2 + \frac{4}{11} v_3 = 0 \quad (1)$$

EXERCISE

Determine the **current** flowing through each resistor and the **voltage** drop across each resistor in the given circuit.



Solution

KVL @ L1

$$-v_0 + v_1 + v_2 = 0$$

$$v_1 + v_2 = v_0$$

$$\underline{v_1 + v_2 = 9V} \quad (2)$$

KVL @ L2

$$\underline{-v_2 + v_3 = 0} \quad (3)$$

EXERCISE

Gaussian Elimination

v_1	v_2	v_3	
-1	$\frac{6}{11}$	$\frac{4}{11}$	0
1	1	0	1
0	-1	1	0

row-echelon form
(a triangular form)

Solution

$$-v_1 + \frac{6}{11}v_2 + \frac{4}{11}v_3 = 0 \quad (1)$$

$$v_1 + v_2 = 1V \quad (2)$$

$$-v_2 + v_3 = 0 \quad (3)$$



EXERCISE

Gaussian Elimination

v_1	v_2	v_3		
<u>-1</u>	<u>$\frac{6}{11}$</u>	<u>$\frac{4}{11}$</u>	0	r_1
<u>1</u>	1	0	9	$r_2 \leftarrow r_1 + r_2$
<u>0</u>	<u>-1</u>	1	0	r_3

v_1	v_2	v_3		
<u>-1</u>	<u>$\frac{6}{11}$</u>	<u>$\frac{4}{11}$</u>	0	r_1
<u>0</u>	<u>$\frac{17}{11}$</u>	<u>$\frac{4}{11}$</u>	9	r_2
<u>0</u>	<u>-1</u>	1	0	$r_3 \leftarrow r_2 + \frac{17}{11} r_3$

Solution

v_1	v_2	v_3		
<u>-1</u>	<u>$\frac{6}{11}$</u>	<u>$\frac{4}{11}$</u>	0	r_1
<u>0</u>	<u>$\frac{17}{11}$</u>	<u>$\frac{4}{11}$</u>	9	r_2
<u>0</u>	<u>0</u>	<u>$\frac{21}{11}$</u>	9	r_3

from r_3

$$\frac{21}{11} v_3 = 9$$

$$v_3 = \frac{9(11)}{21}$$

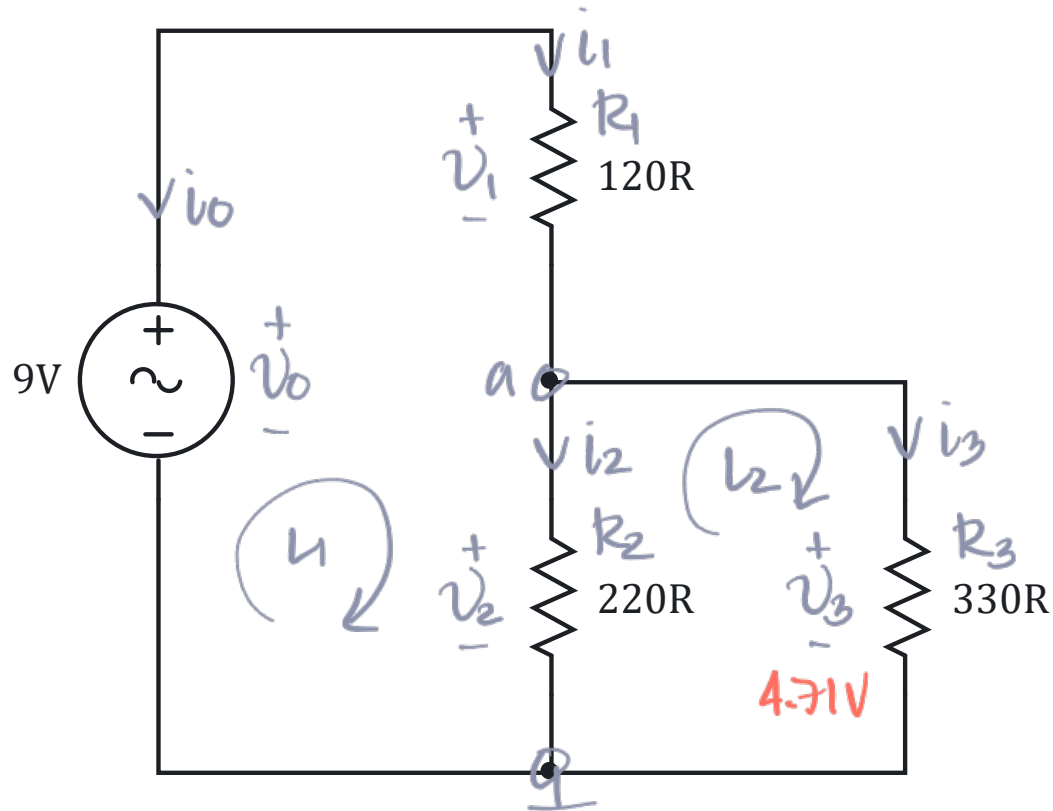
$$v_3 = 4.71 \text{ V}$$

ans



EXERCISE

Determine the **current** flowing through each resistor and the **voltage** drop across each resistor in the given circuit.



Solution

$$-v_2 + v_3 = 0 \quad (3)$$

$$v_2 = v_3$$

$$v_2 = 4.71V$$

ans

$$v_1 + v_2 = 9V \quad (2)$$

$$v_1 = 9 - v_2$$

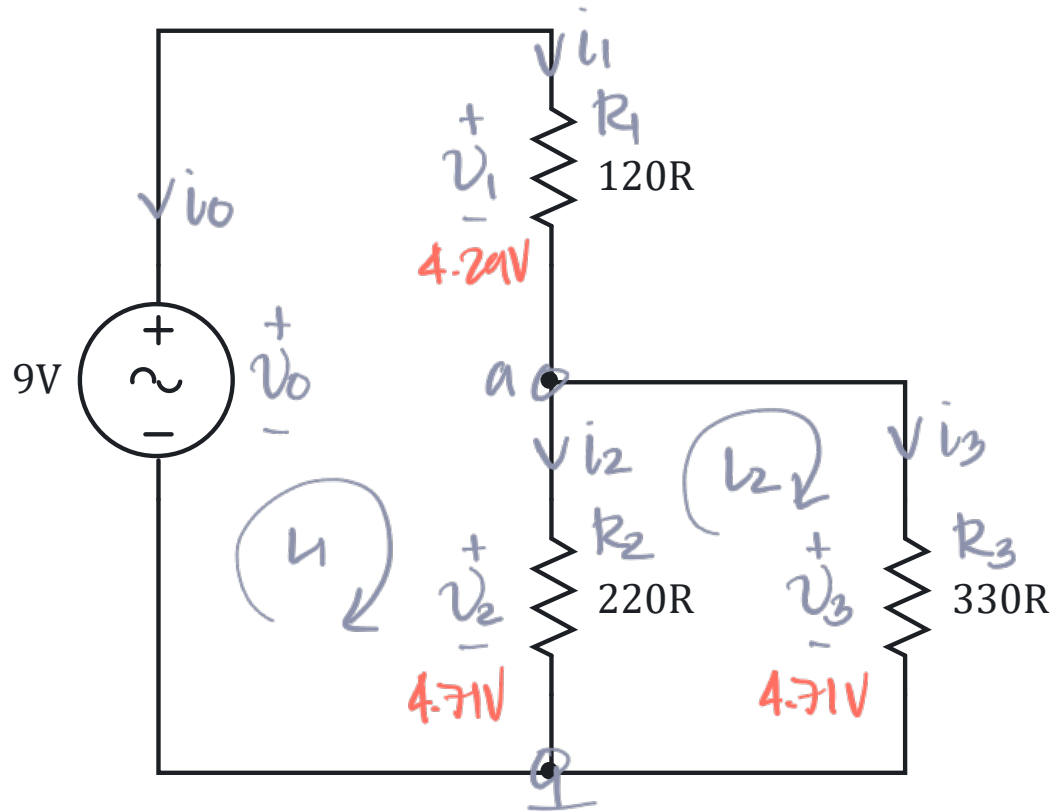
$$v_1 = 9 - 4.71$$

$$v_1 = 4.29V$$

ans

EXERCISE

Determine the **current** flowing through each resistor and the **voltage** drop across each resistor in the given circuit.



Solution

$$i_1 = \frac{V_1}{R_1}$$

$$i_1 = \frac{4.29}{120}$$

$$i_1 = 35.75 \text{ mA}$$

ans

$$i_3 = \frac{V_3}{R_3}$$

$$i_3 = \frac{4.71}{330}$$

$$i_3 = 14.27 \text{ mA}$$

ans

$$i_2 = \frac{V_2}{R_2}$$

$$i_2 = \frac{4.71}{220}$$

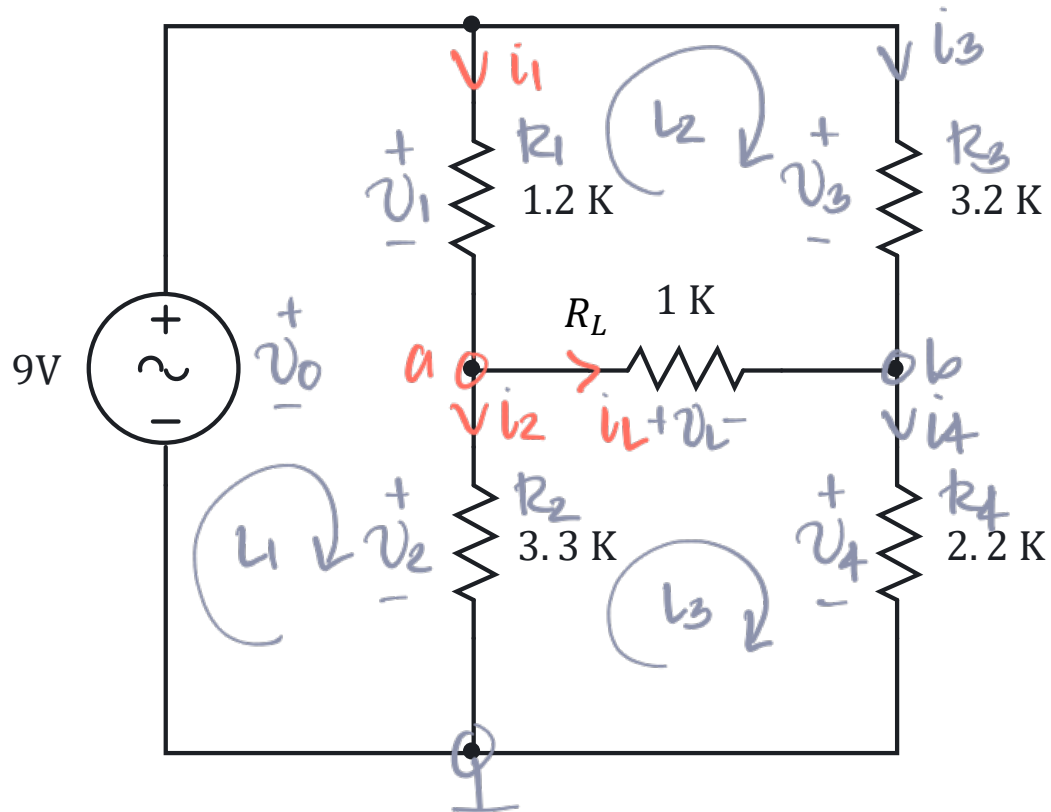
$$i_2 = 21.41 \text{ mA}$$

ans



EXERCISE

Determine the load R_L voltage and current of the given circuit.



Solution

KCL @ a

$$-i_1 + i_2 + i_L = 0$$

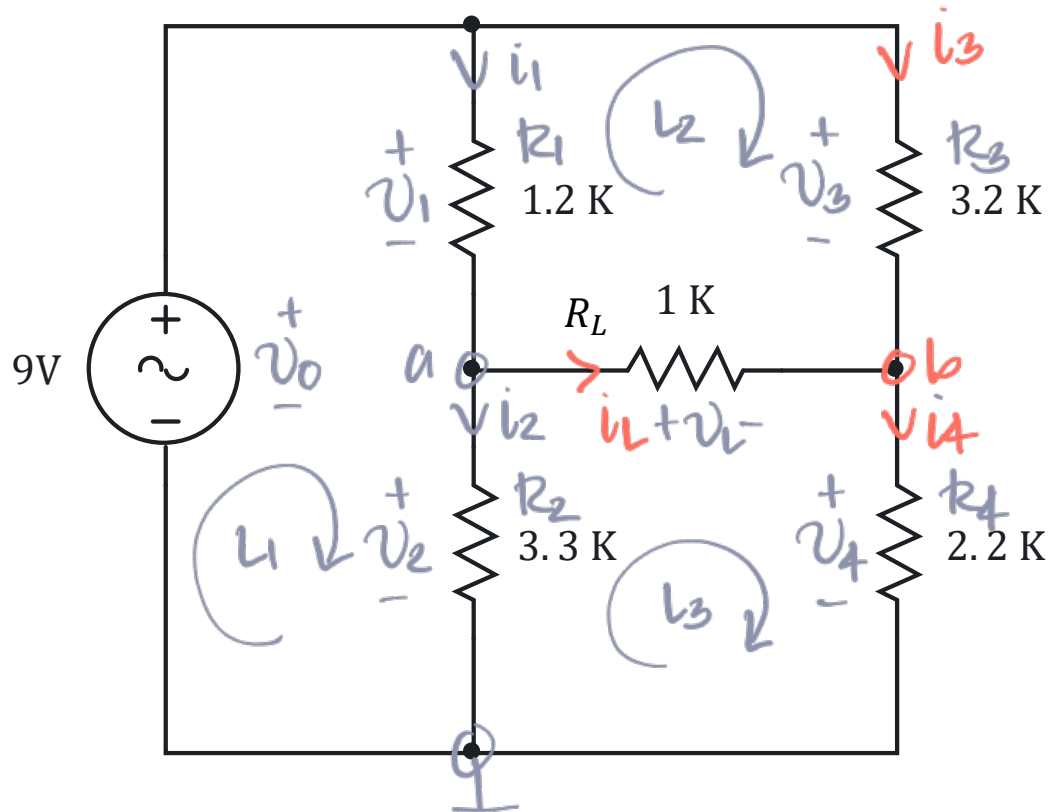
$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_L}{R_L} = 0$$

$$1.2k \left[-\frac{v_1}{1.2k} + \frac{v_2}{3.3k} + \frac{v_L}{1k} \right] = 0$$

$$-v_1 + \frac{4}{11}v_2 + 1.2v_L = 0 \quad (1)$$

EXERCISE

Determine the load R_L voltage and current of the given circuit.



Solution

KCL @ b

$$-i_L - i_3 + i_4 = 0$$

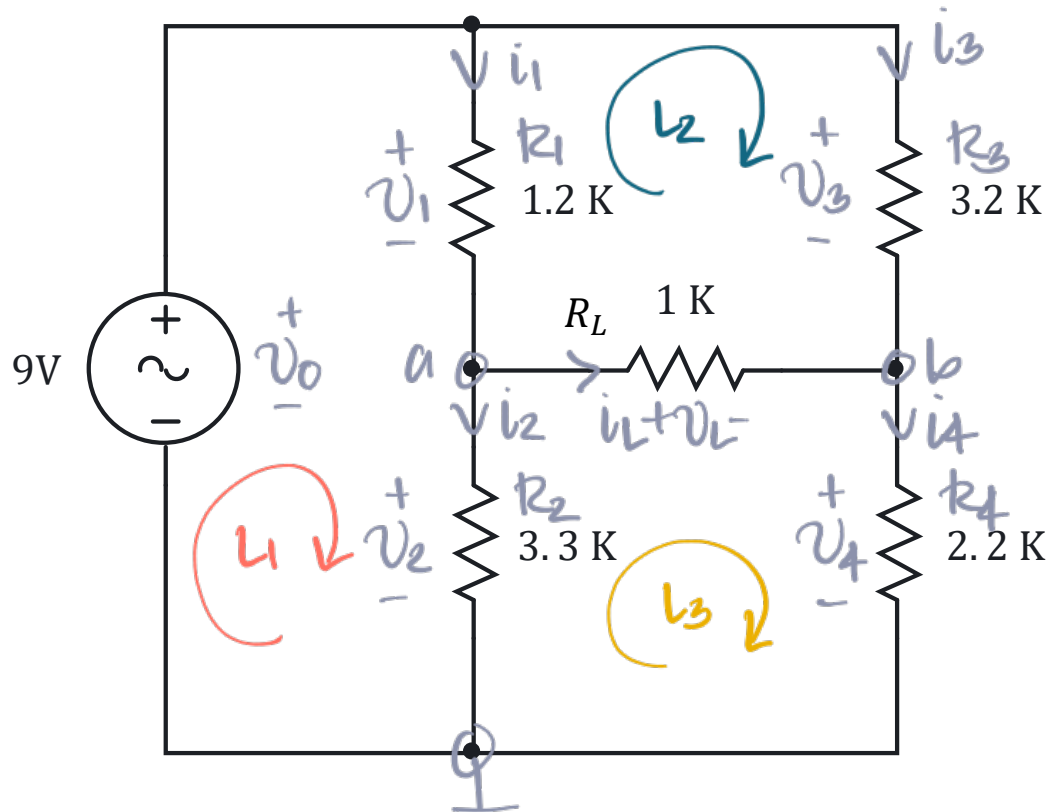
$$-\frac{v_L}{R_L} - \frac{v_3}{R_3} + \frac{v_4}{R_4} = 0$$

$$\left[-\frac{v_L}{1k} - \frac{v_3}{3.2k} + \frac{v_4}{2.2k} = 0 \right]$$

$$-v_3 + \frac{16}{11} v_4 - 3.2 v_L = 0 \quad (2)$$

EXERCISE

Determine the load R_L voltage and current of the given circuit.



Solution

KVL@ L1

$$-\cancel{v_0}^9 + v_1 + v_2 = 0$$

$$\underline{v_1 + v_2 = 9 \quad (3)}$$

KVL@ L2

$$\underline{-v_1 + v_3 - v_L = 0 \quad (4)}$$

KVL@ L3

$$\underline{-v_2 + v_4 + v_L = 0 \quad (5)}$$



EXERCISE

Gaussian Elimination

v_1	v_2	v_3	v_4	v_L		
-1	$\frac{4}{11}$	0	0	1.2	0	r_1
0	0	-1	$\frac{16}{11}$	-3.2	0	r_2
1	1	0	0	0	9	r_3
-1	0	1	0	-1	0	r_4
0	-1	0	1	1	0	r_5

Solution

$$-v_1 + \frac{4}{11}v_2 + 1.2v_L = 0 \quad (1)$$

$$-v_3 + \frac{16}{11}v_4 - 3.2v_L = 0 \quad (2)$$

$$v_1 + v_2 = 9 \quad (3)$$

$$-v_1 + v_3 - v_L = 0 \quad (4)$$

$$-v_2 + v_4 + v_L = 0 \quad (5)$$



EXERCISE

Gaussian Elimination

v_1	v_2	v_3	v_4	v_L	
-1	$\frac{4}{11}$	0	0	1.2	0
0	0	-1	$\frac{16}{11}$	-3.2	0
1	1	0	0	0	1
-1	0	1	0	-1	0
0	-1	0	1	1	0

Solution

v_1	v_2	v_3	v_4	v_L	
-1	$\frac{4}{11}$	0	0	1.2	0
0	-1	0	1	1	0
0	0	-1	$\frac{16}{11}$	-3.2	0
-1	0	1	0	-1	0
1	1	0	0	0	1

Rearrange rows

EXERCISE

Gaussian Elimination

v_1	v_2	v_3	v_4	v_L		
-1	$\frac{4}{11}$	0	0	1.2	0	r_1
<u>0</u>	-1	0	1	1	0	r_2
<u>0</u>	<u>0</u>	-1	$\frac{16}{11}$	-3.2	0	r_3
<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	-1	9	r_4
<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	0	9	r_5

Solution

v_1	v_2	v_3	v_4	v_L		
-1	$\frac{4}{11}$	0	0	1.2	0	r_1
<u>0</u>	-1	0	1	1	0	r_2
<u>0</u>	<u>0</u>	-1	$\frac{16}{11}$	-3.2	0	r_3
<u>-1</u>	<u>0</u>	<u>1</u>	<u>0</u>	-1	0	r_4
<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	0	9	r_5

$$r_4 \leftarrow r_4 + r_5$$



EXERCISE

Gaussian Elimination

v_1	v_2	v_3	v_4	v_L		
-1	$\frac{4}{11}$	0	0	1.2	0	r_1
<u>0</u>	-1	0	1	1	0	r_2
<u>0</u>	<u>0</u>	-1	$\frac{16}{11}$	-3.2	0	r_3
<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	-1	9	r_4
<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	0	9	r_5

$$r_4 \leftarrow r_2 + r_4$$

Solution

v_1	v_2	v_3	v_4	v_L		
-1	$\frac{4}{11}$	0	0	1.2	0	r_1
<u>0</u>	-1	0	1	1	0	r_2
<u>0</u>	<u>0</u>	-1	$\frac{16}{11}$	-3.2	0	r_3
<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	0	9	r_4
<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	0	9	r_5



EXERCISE

Gaussian Elimination

v_1	v_2	v_3	v_4	v_L		
-1	$\frac{4}{11}$	0	0	1.2	0	r_1
<u>0</u>	-1	0	1	1	0	r_2
<u>0</u>	<u>0</u>	-1	$\frac{16}{11}$	-3.2	0	r_3
<u>0</u>	<u>0</u>	<u>0</u>	$\frac{27}{11}$	-3.2	9	r_4
<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	0	9	r_5

Solution

v_1	v_2	v_3	v_4	v_L		
-1	$\frac{4}{11}$	0	0	1.2	0	r_1
<u>0</u>	-1	0	1	1	0	r_2
<u>0</u>	<u>0</u>	-1	$\frac{16}{11}$	-3.2	0	r_3
<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	0	9	r_4
<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	0	9	r_5

$$r_4 \leftarrow r_3 + r_4$$

EXERCISE

Gaussian Elimination

v_1	v_2	v_3	v_4	v_L		
-1	$\frac{4}{11}$	0	0	1.2	0	r_1
<u>0</u>	-1	0	1	1	0	r_2
<u>0</u>	<u>0</u>	-1	$\frac{16}{11}$	-3.2	0	r_3
<u>0</u>	<u>0</u>	<u>0</u>	$\frac{27}{11}$	-3.2	9	r_4
<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>0</u>	9	r_5

$$r_5 \leftarrow r_1 + r_5$$

Solution

v_1	v_2	v_3	v_4	v_L		
-1	$\frac{4}{11}$	0	0	1.2	0	r_1
<u>0</u>	-1	0	1	1	0	r_2
<u>0</u>	<u>0</u>	-1	$\frac{16}{11}$	-3.2	0	r_3
<u>0</u>	<u>0</u>	<u>0</u>	$\frac{27}{11}$	-3.2	9	r_4
<u>0</u>	$\frac{15}{11}$	<u>0</u>	<u>0</u>	1.2	9	r_5



EXERCISE

Gaussian Elimination

v_1	v_2	v_3	v_4	v_L		
-1	$\frac{4}{11}$	0	0	1.2	0	r_1
<u>0</u>	-1	0	1	1	0	r_2
<u>0</u>	<u>0</u>	-1	$\frac{16}{11}$	-3.2	0	r_3
<u>0</u>	<u>0</u>	<u>0</u>	$\frac{27}{11}$	-3.2	9	r_4
<u>0</u>	<u>0</u>	<u>0</u>	$\frac{15}{11}$	$\frac{141}{11}$	9	r_5

Solution

v_1	v_2	v_3	v_4	v_L		
-1	$\frac{4}{11}$	0	0	1.2	0	r_1
<u>0</u>	-1	0	1	1	0	r_2
<u>0</u>	<u>0</u>	-1	$\frac{16}{11}$	-3.2	0	r_3
<u>0</u>	<u>0</u>	<u>0</u>	$\frac{27}{11}$	-3.2	9	r_4
<u>0</u>	$\frac{15}{11}$	<u>0</u>	<u>0</u>	1.2	9	r_5

$$r_5 \leftarrow \frac{15}{11} r_2 + r_5$$



EXERCISE

Gaussian Elimination

v_1	v_2	v_3	v_4	v_L		
-1	$\frac{4}{11}$	0	0	1.2	0	r_1
<u>0</u>	-1	0	1	1	0	r_2
<u>0</u>	<u>0</u>	-1	$\frac{16}{11}$	-3.2	0	r_3
<u>0</u>	<u>0</u>	<u>0</u>	$\frac{27}{11}$	-3.2	9	r_4
<u>0</u>	<u>0</u>	<u>0</u>	$\frac{15}{11}$	$\frac{141}{11}$	9	r_5

$$r_5 \leftarrow \frac{15}{11} r_4 - \frac{27}{11} r_5$$

Solution

v_1	v_2	v_3	v_4	v_L		
-1	$\frac{4}{11}$	0	0	1.2	0	r_1
<u>0</u>	-1	0	1	1	0	r_2
<u>0</u>	<u>0</u>	-1	$\frac{16}{11}$	-3.2	0	r_3
<u>0</u>	<u>0</u>	<u>0</u>	$\frac{27}{11}$	-3.2	9	r_4
<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>	-10.66	-9.82	r_5

EXERCISE

from v_5

$$\frac{-10.66 v_L}{-10.66} = \frac{-9.82}{-10.66}$$

$$v_L = 921.20 \text{ mV}$$

ans

Solution

v_1	v_2	v_3	v_4	v_L		
-1	$\frac{4}{11}$	0	0	1.2	0	v_1
<u>0</u>	-1	0	1	1	0	v_2
<u>0</u>	<u>0</u>	-1	$\frac{16}{11}$	-3.2	0	v_3
<u>0</u>	<u>0</u>	<u>0</u>	$\frac{27}{11}$	-3.2	9	v_4
<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>	-10.66	-9.82	v_5

LABORATORY

